National Institute of Diabetes and Digestive and Kidney Diseases

Anemia in Chronic Kidney Disease

On this page:

What is anemia?

Anemia NIH is a condition in which your blood has a lower-than-normal amount of red blood cells or hemoglobin. Hemoglobin is the iron NIH-rich protein that allows red blood cells to carry oxygen from your lungs to the rest of your body. With fewer red blood cells or less hemoglobin, your tissues and organs —such as your heart and brain—may not get enough oxygen to work properly.

How is anemia related to chronic kidney disease?

Anemia is a common complication of chronic kidney disease (CKD). CKD means your kidneys are damaged and can't filter blood the way they should. This damage can cause wastes and fluid to build up in your body. CKD can also cause other health problems.

Anemia is less common in early kidney disease, and it often gets worse as kidney disease progresses and more kidney function is lost.

Does anemia in CKD have another name?

Anemia in CKD is also called anemia of renal disease.

How common is anemia in CKD?

Anemia is common in people with CKD, especially among people with more advanced kidney disease. More than 37 million American adults may have CKD,[1] and it is estimated that more than 1 out of every 7 people with kidney disease have anemia.[2]

Most people who have kidney failure—when kidney damage is so advanced that less than 15 percent of the kidney is working normally—also have anemia.[3]

Who is more likely to have anemia in CKD?

Your risk for anemia increases as your kidney disease gets worse.

People with CKD who also have diabetes are at greater risk for anemia, tend to develop anemia earlier, and often have more severe anemia than people with CKD who don't have diabetes.[4] People older than 60 are also more likely to have anemia with CKD.[5]

What are the complications of anemia in someone with CKD?

In people with CKD, severe anemia can increase the chance of developing heart problems because the heart is getting less oxygen than normal and is working harder to pump enough red blood cells to organs and tissues. People with CKD and anemia may also be at an increased risk for complications due to strokes NIH
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What are the symptoms of anemia in someone with CKD?

Anemia related to CKD typically develops slowly and may cause few or no symptoms in early kidney disease.

Symptoms of anemia in CKD may include

- fatigue or tiredness
- shortness of breath
- unusually pale skin
- weakness
- body aches

- chest pain
- dizziness
- fainting
- fast or irregular heartbeat
- headaches
- sleep problems
- trouble concentrating

Fatigue can be a symptom of anemia. Talk with your health care professional if you are feeling unusually tired or have other symptoms of anemia.

Seek care right away

Seek emergency medical care by calling 911 if you have chest pain that won’t go away.

If you have difficulty breathing or shortness of breath, seek immediate medical care.

What causes anemia in CKD?

Anemia in people with CKD often has more than one cause.

When your kidneys are damaged, they produce less erythropoietin (EPO), a hormone that signals your bone marrow—the spongy tissue inside most of your bones—to make red blood cells. With less EPO, your body makes fewer red blood cells, and less oxygen is delivered to your organs and tissues.

In addition to your body making fewer red blood cells, the red blood cells of people with anemia and CKD tend to live in the bloodstream for a shorter time than normal, causing the blood cells to die faster than they can be replaced.

People with anemia and CKD may have low levels of nutrients, such as iron, vitamin B12 NIH, and folate NIH, that are needed to make healthy red blood cells.

Other causes of anemia related to CKD include

- blood loss, particularly if you are treated with dialysis for kidney failure
- infection
- inflammation
- malnutrition, a condition that occurs when the body doesn't get enough nutrients

How do health care professionals diagnose anemia in CKD?

Health care professionals use your medical history, a physical exam, and blood tests to diagnose anemia in CKD.

Medical history

Your health care professional will record your medical history and may ask about

- your symptoms
- current and past medical conditions
- prescription and over-the-counter medicines you take
- your family history

Physical exam

During a physical exam, your health care professional may

- check your blood pressure
- check your heart rate
- examine your body, including checking for changes in skin color, rashes, or bruising

Blood tests

Health care professionals use blood tests NIH to check for signs of anemia or other health problems. Your health care professional will take a blood sample from you and send the sample to a lab to test.

Blood count tests NIH can check many parts and features of your blood, including the

- number of red blood cells
- average size of red blood cells
- amount of hemoglobin in your blood and in your red blood cells
- number of developing red blood cells, called reticulocytes, in your blood

Some of these blood count tests and others may be combined in a test called a complete blood count NIH, or CBC.

Your health care professional may also use blood tests to check the amount of iron in your blood and stored in your body. These tests may measure

- ferritin, the protein that stores iron in your body's cells
- transferrin, a protein in your blood that carries iron

Health care professionals also sometimes use blood tests to check for low levels of folate and vitamin B12.

If blood test results suggest you have anemia but the cause is unknown, your health care professional may perform additional tests to look for the cause or may refer you to a hematologist, a health care professional who treats blood disorders.

How do health care professionals treat anemia in CKD?

Health care professionals first treat any underlying conditions that may be causing the anemia, such as an iron or vitamin deficiency. If your anemia is mild and you have few symptoms, you may not need treatment at first.

Treatments for anemia may ease your symptoms and improve your quality of life.

Your health care professional may refer you to a hematologist or a nephrologist, a health care professional who treats people with kidney problems or related conditions.

Iron

If you don't have enough iron in your body, your health care professional may prescribe iron supplements, either as a pill or intravenous (IV) infusion. If you're on dialysis, you may be given an IV iron supplement during your dialysis treatment. Iron supplements help your body make healthy red blood cells.

Vitamins

Your health care professional may ask you to take vitamin supplements such as vitamin B12 or folate—both needed to make healthy blood cells—if your body doesn't have enough of these vitamins.

Medicines

Your health care professional may prescribe an erythropoiesis-stimulating agent (ESA) to treat your anemia. ESAs send a signal to your bone marrow to make more red blood cells.

If you're on hemodialysis, you may receive IV or subcutaneous ESAs during your dialysis treatments. If you are on peritoneal dialysis or do not receive dialysis, your health care professional may give ESAs as shots and may teach you how to give yourself these shots at home.

Your health care professional may prescribe iron supplements to help ESAs work better or to reduce the amount of ESAs you need.

ESAs may ease your symptoms and help you avoid blood transfusions NIH, but the treatment is not right for everyone with CKD and anemia. Talk with your health care professional about the risks and benefits of ESAs and if the medicine is right for you.

Blood transfusions

In some cases, health care professionals may use blood transfusions to treat severe anemia in CKD. A blood transfusion can quickly increase the number of red blood cells in your body and temporarily relieve the symptoms of anemia.

Health care professionals may limit or avoid blood transfusions because they can sometimes lead to other health problems, including

- the body may develop antibodies over time that damage or destroy the donor blood cells and may delay or reduce the possibility of a future kidney transplant
- iron NIH from transfused red blood cells can build up in the body and damage organs, called iron overload or hemochromatosis

Talk with your health care professional about your treatment options and the benefits and risks of each treatment.

Can I prevent anemia in CKD?

You may not be able to prevent anemia, but managing your kidney disease may help delay anemia or prevent anemia from getting worse.

How does eating, diet, and nutrition affect anemia in CKD?

You may need to change what you eat to manage your anemia and CKD. Work with your health care professional or a registered dietitian to develop a meal plan that includes foods that you enjoy eating while maintaining your kidney health and managing your anemia.

If your body doesn't have enough iron, vitamin B12, or folate, your health care professional or a dietitian may suggest that you add more foods with these nutrients to your diet. However, some of these foods have high amounts of protein, sodium, or phosphorus, which people with CKD may need to limit. Talk with your health care professional or a dietitian before making any changes to your diet.

Clinical Trials for Anemia in CKD

The NIDDK conducts and supports clinical trials in many diseases and conditions, including kidney diseases. The trials look to find new ways to prevent, detect, or treat disease and improve quality of life.

What are clinical trials for anemia in CKD?

Clinical trials—and other types of clinical studies NIH—are part of medical research and involve people like you. When you volunteer to take part in a clinical study, you help health care professionals and researchers learn more about disease, and improve health care for people in the future.

Researchers are studying many aspects of anemia in CKD, such as

- the relationship between the gut microbiome and anemia in CKD
- new treatments for anemia in people with CKD
- improving the safety and efficacy of treatments for anemia in people with CKD
- optimal hemoglobin target levels for people with CKD

Find out if clinical studies are right for you NIH.

Watch a video of NIDDK Director Dr. Griffin P. Rodgers explaining the importance of participating in clinical trials.

Why Should I Join a Clinical Trial?

National Institute of Diabetes and Digestive and Kidney Diseases (NIDDK)

From a US national research authority

Watch on

What clinical studies for anemia in CKD are looking for participants?

You can view a filtered list of clinical studies on anemia in CKD that are open and recruiting at ClinicalTrials.gov NIH. You can expand or narrow the list to include clinical studies from industry, universities, and individuals; however, the NIH does not review these studies and cannot ensure they are safe. Always talk with your health care professional before you participate in a clinical study.

References

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